₹50,000 crore.

lost every year to COD returns.

COD returns

3 IN 10

COD parcels bounce.

THIS IS HOW YOU STOP THEM—

BEFORE THE COURIER LEAVES.

RTO TRUST LAYER

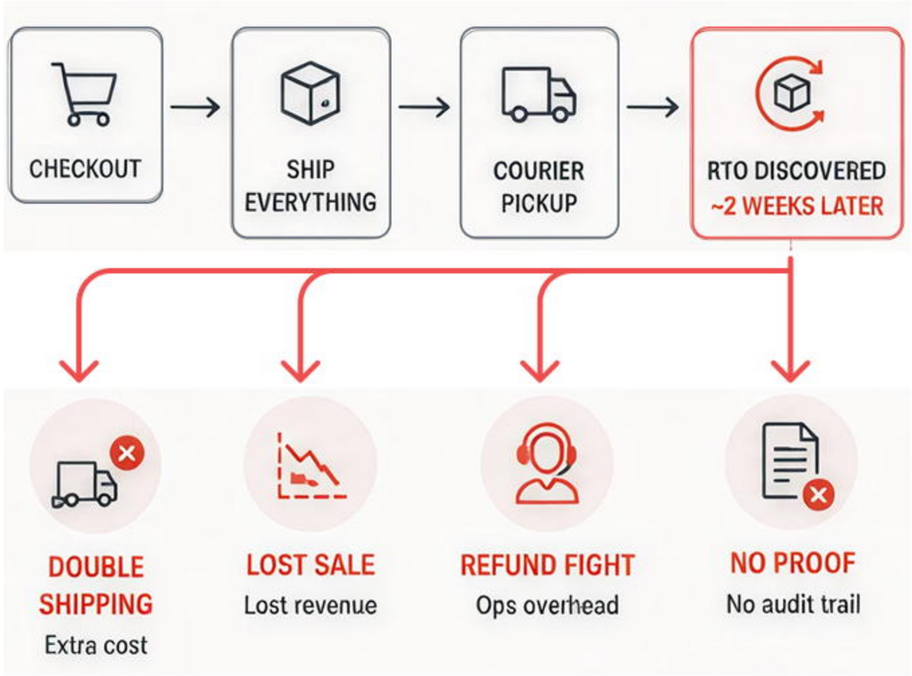
REAL-TIME COD RISK DECISION + AUDIT

PROBLEM — DECIDE BEFORE THE COURIER LEAVES

MOVE THE DECISION | FROM AFTER SHIPMENT → TO AT CHECKOUT

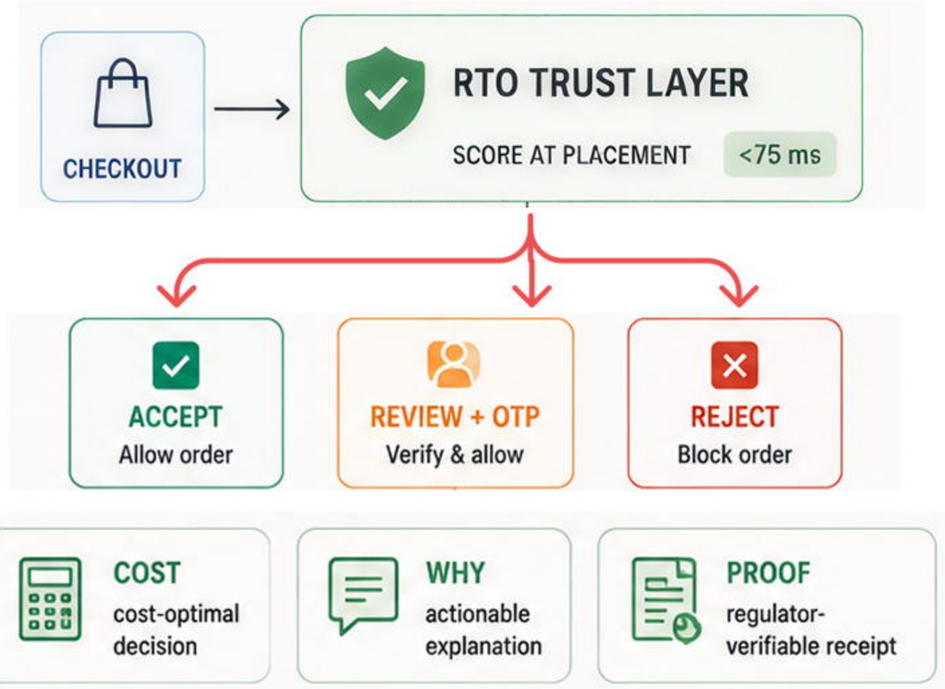
TODAY

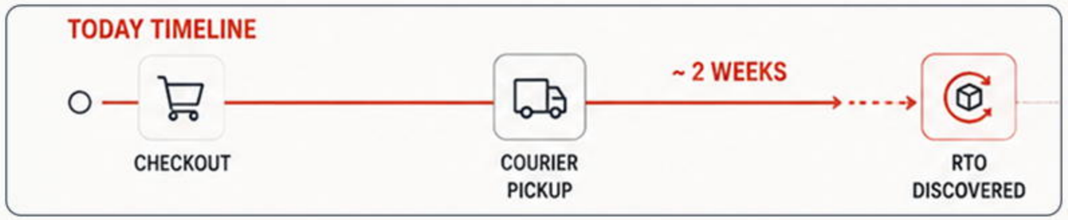
DISCOVERY AFTER SHIPMENT



PROOF

DECISION AT CHECKOUT

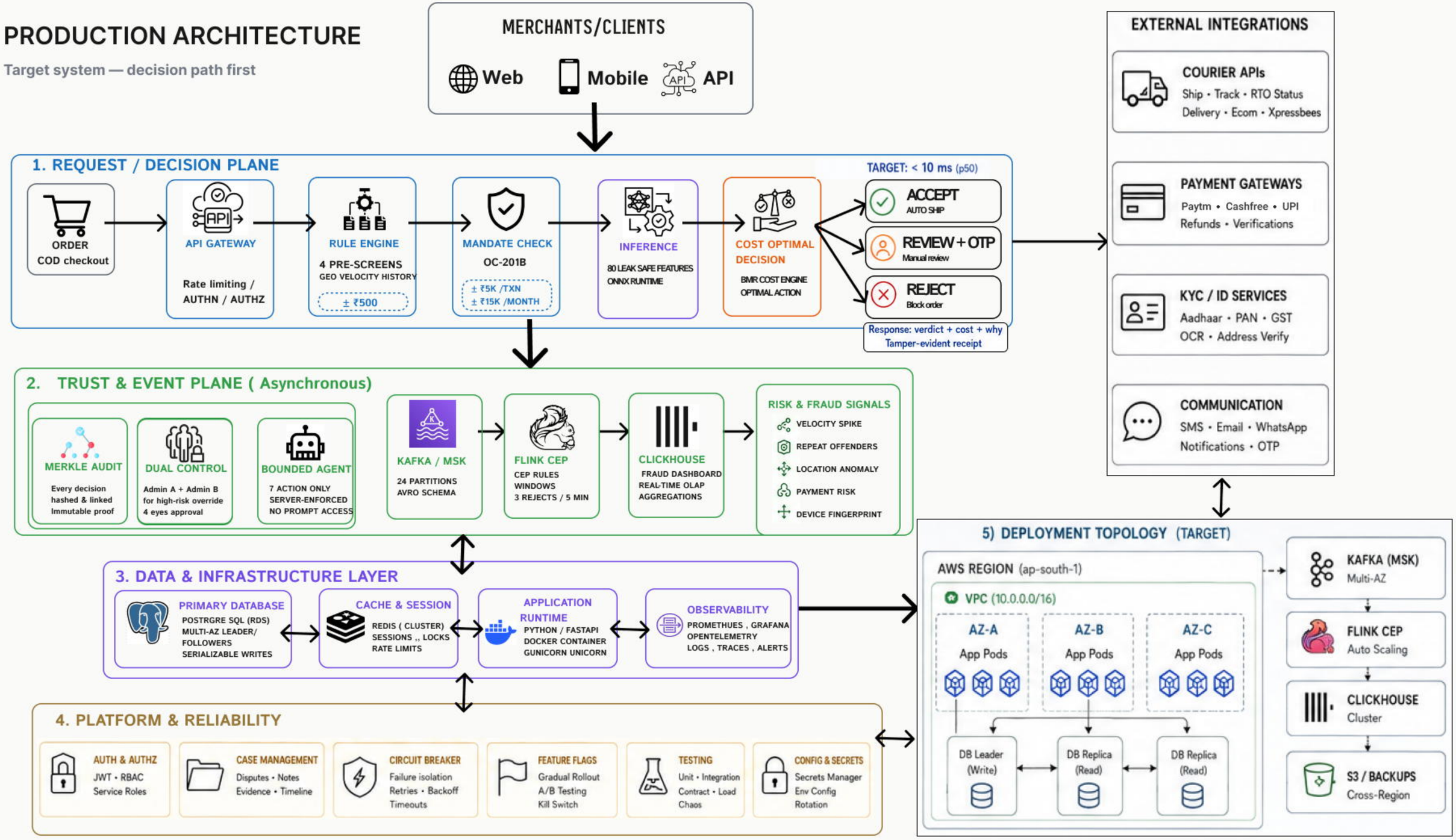






PRODUCTION ARCHITECTURE

Target system — decision path first



INFERENCE — 79 LEAK-SAFE FEATURES

Feature construction → ensemble → serving → cost decision

1) FEATURE ENGINEERING (Leak-Safe)

RAW ORDER EVENT

- order_id
- timestamp
- pincode
- qty
- category
- price
- ...

79 LEAK-SAFE FEATURES

5 Major Families

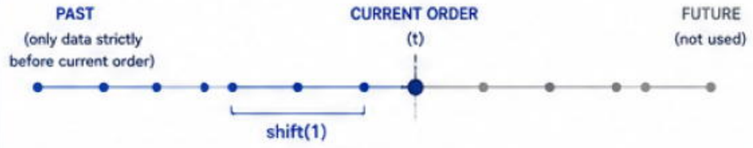
BEHAVIOURAL / RATES	expanding mean, rate, return rate, RTO rate, shifted stats (t-1)	23 features
GEOGRAPHY	city, pincode, region, tier, state aggregates, distance bands	18 features
ORDER SIGNALS	quantity, category, amount buckets, is_qty_zero, COD, etc.	16 features
HISTORY SIGNALS	cust. return rate (t-1), freq. band, tenure band, past behaviour	14 features
CATEGORICAL EMBED / FLAGS	promo, payment mode, delivery type, channel, seasonality flags	8 features

FEATURE PROCESSING

- Expanding Rates with shift(1)
- m-Estimate Smoothing (m = 20)
- City / Pincode Smoothing
- Imputation / Encoding (Leak-Safe)

OUTPUT 79-DIM VECTOR

→ LEAKAGE PREVENTION (Strict)



NO FUTURE INFORMATION USED

2) MODELING & ENSEMBLE (Training)

XGBOOST (Champion)

weight (w_1) = 0.936

- depth = 7
- scale_pos_weight = 9.65
- learning_rate = 0.058
- subsample = 0.85

Output: p_1

HISTGB

weight (w_2) = 0.103

- max_iter = 400
- max_leaf_nodes = 33
- learning_rate = 0.05
- l2_regularization = 1.0

Output: p_2

OPTUNA-WEIGHTED ENSEMBLE

$$p_{ens} = \frac{\sum_i w_i \cdot p_i}{\sum_i w_i}$$

Optuna search on time-split validation
weights are non-negative and sum to 1

CHAMPION ENSEMBLE SCORE p_{ens}

TRAINING / MODEL BUILD

3) SERVING / INFERENCE

MODEL ARTIFACT

model.onnx
opset: 17
input: FloatTensor
shape: [None, 79]
size: 48 KB

ONNX RUNTIME

CPUExecutionProvider

~ 1.6 µs / row
(single core)

RTO RISK SCORE

0 (low risk) — 1 (high risk)

SERVING / INFERENCE

4) COST OPTIMAL DECISION

BAHNSEN COST ENGINE

(Cost-Minimization)

Compute expected cost for each action using cost matrix and risk score p_{ens}

EXPECTED COST (₹)	
ACCEPT	₹248
REVIEW + OTP	₹99
REJECT	₹980

E[action] = C(action | x)
choose action with minimum expected cost

SELECTED ACTION

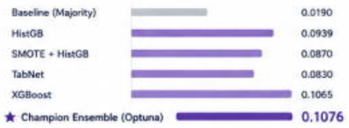
REVIEW + OTP
(min expected cost)

ACCEPT	REVIEW + OTP	REJECT
₹248	₹99	₹980

FEATURE ENGINEERING HIGHLIGHTS

- Expanding stats use only prior data (shift 1)
- m-estimate smoothing (m = 20) for sparse cities
- City / pincode aggregates built on past window
- All categorical encodings are leak-safe
- 79 features total → 79-dim dense vector

MODEL SELECTION (Time-Split PR-AUC)



DATA / EVAL CONTEXT

- ~1.3L Orders — Amazon India / Public datasets
- Strict Time Split — Train → Valid → Test (no leakage)
- No User History (for this evaluation)
- Base Rate (RTO) ≈ 1.9% (test set)

NOTES

- Inference runs entirely in-memory.
- ONNX Runtime ensures low latency and portability.
- Decision layer is independent of model and thresholds.
- All numbers are from time-split test evaluation.
- Costs are illustrative and configurable.

Data / Flow
Training Flow
Artifact / Deploy
Decision Flow
Data / Feature Store
Model Artifact

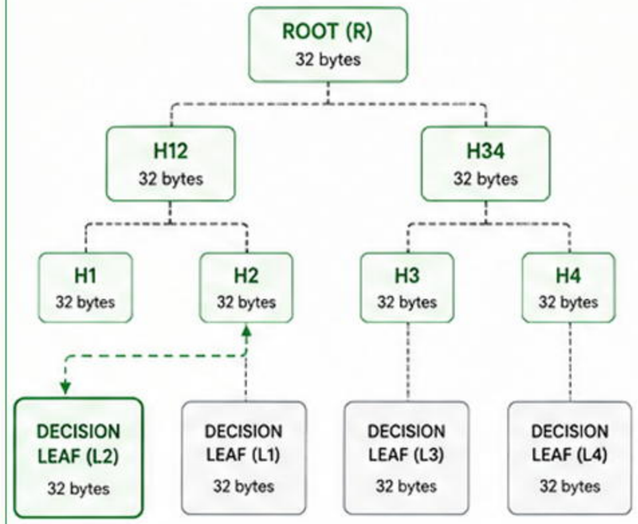
TRUST CORE — 3 DIFFERENTIATORS

Not just a model with an API in front of it.

DECISION
Accept / Review / Reject

01 MERKLE AUDIT CHAIN

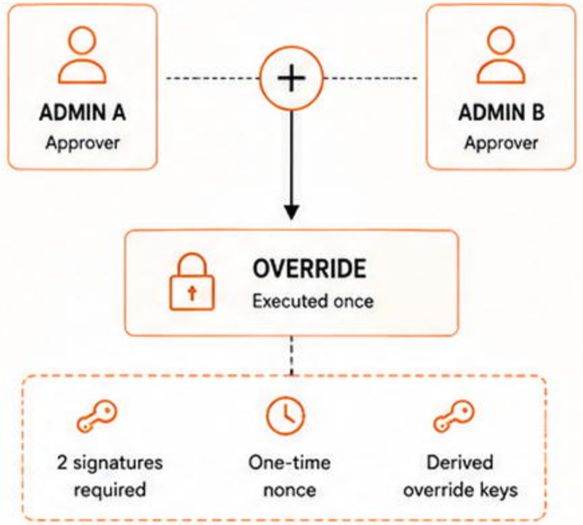
Every decision is immutably hashed and linked to a root.



Tamper one record → proof breaks.
Auditor can verify in O(log n).

02 DUAL CONTROL OVERRIDE

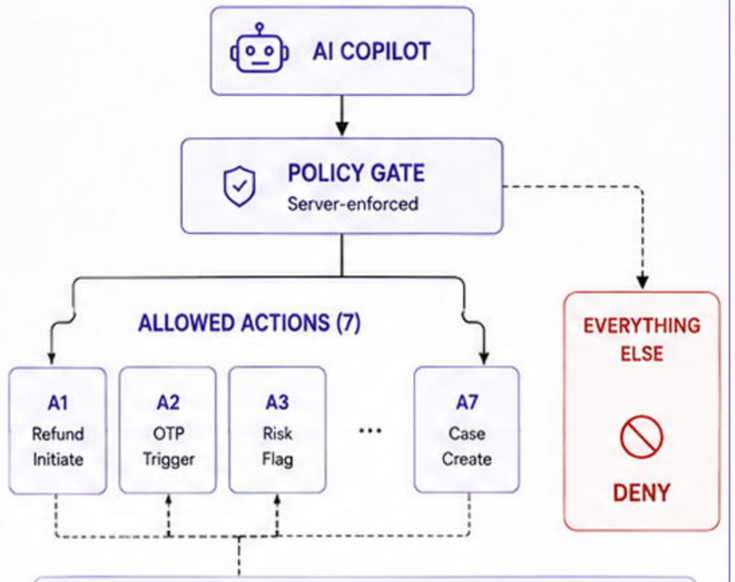
No single admin can reverse a decision.
Two approvals + one-time cryptographic keys.



Override is safe, accountable, and cryptographically constrained.

03 BOUNDED AGENT (AI COPILOT)

AI can act, but only within a server-enforced allowlist of 7 actions.



Allowlist enforced in code.
Everything else is denied by default.

TRUST CORE OUTPUT

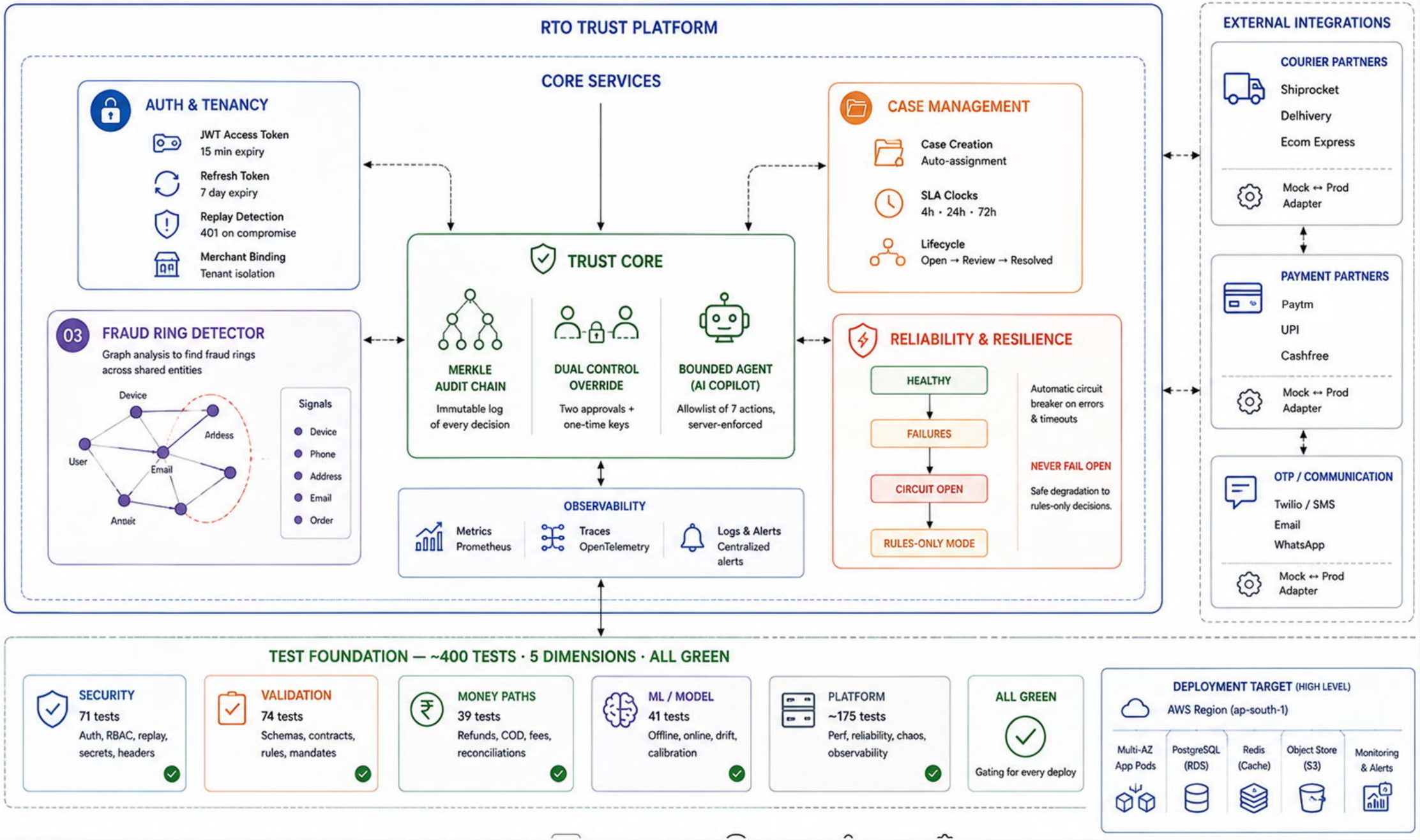
- Tamper-evident receipt (Merkle proof)
- Override record (if any)
- Agent action log (with policy ID)

AUDIT & RECONCILIATION

Any auditor can verify decision integrity end-to-end.

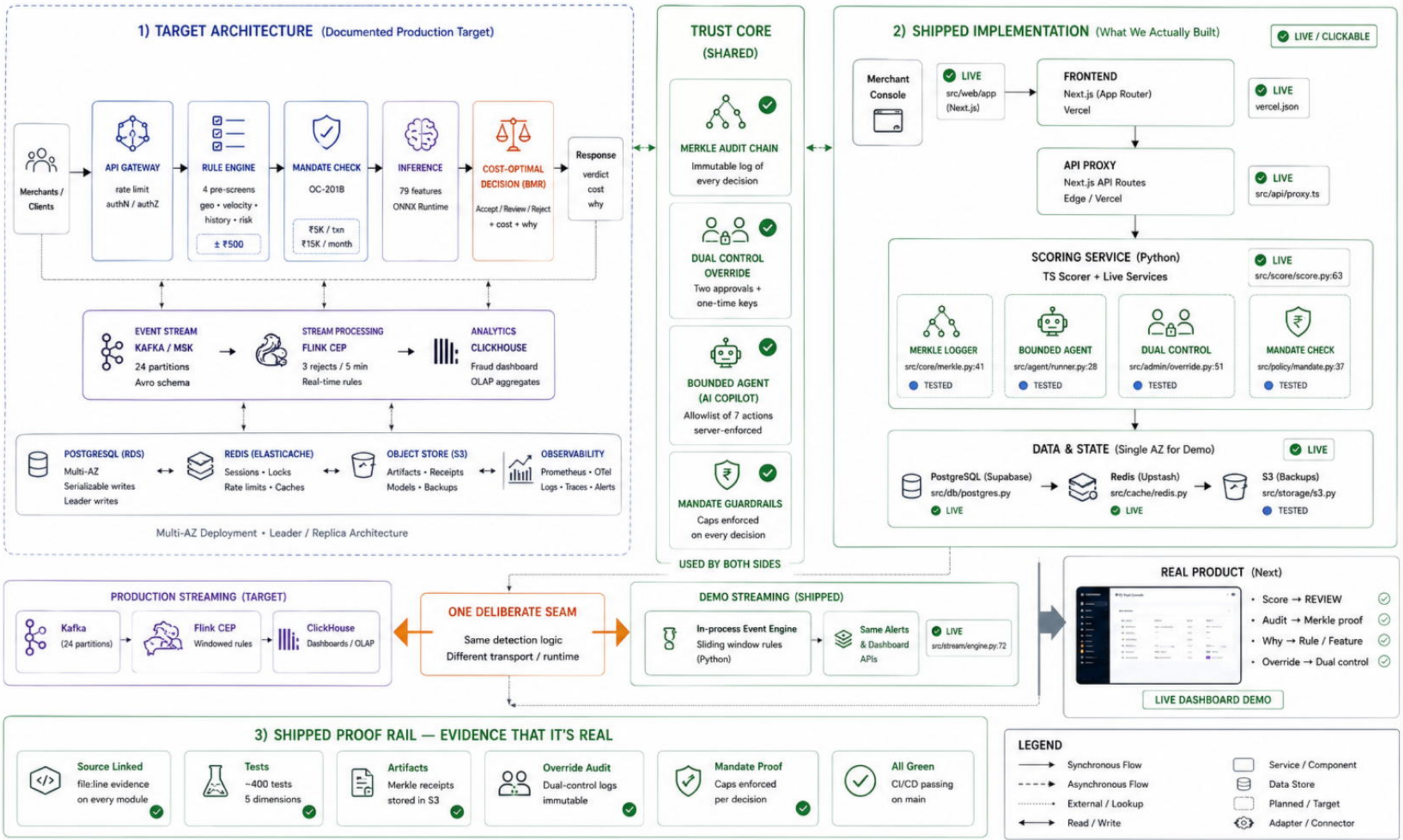
OPERATIONAL BACKBONE

Everything around the trust core that makes the system operable.



THE PIVOT — TARGET vs SHIPPED

What we planned → what is live today



RESULTS — HONEST, TIME-SPLIT, NO LEAKAGE

Measured signal first. Independent cross-check second. Gaps before you ask.

1) AMAZON INDIA / PUBLIC DATA

~1.3L ORDERS · REAL ORDERS

CLASS IMBALANCE (BASE RATE)

Normal Orders (≈98.1%)

RTO / Return (≈1.9%)

Base Rate ≈ 1.9%

CHAMPION SIGNAL

5.7×

base-rate lift

PR-AUC	0.1076
ROC-AUC	0.8934

STRICT TIME SPLIT — NO LEAKAGE

TRAIN Jan → VALIDATION Jul → TEST Dec

NO LEAKAGE ACROSS TIME

SAME PIPELINE

FEATURES → MODELS → WEIGHTED ENSEMBLE → RTO PROBABILITY

TIME-SPLIT PIPELINE
SAME MODEL FAMILY

2) OLIST — REAL CUSTOMER HISTORY

REAL DATA WITH PER-CUSTOMER HISTORY

HISTORY = SIGNAL

Customer A
Customer B
Customer C
...
Customer N

→ History Features

Real customer history unlocks signal we can't see in single-order data.

CROSS-CHECK SIGNAL

32×

base-rate lift

PR-AUC	0.3950

INDEPENDENT VALIDATION — SAME PIPELINE

Stronger signal confirms it's not overfitting.

MODEL COMPARISON (PR-AUC — TIME-SPLIT)

01	HistGB (Champion) ★	0.1076
02	XGBoost	0.0980
03	SMOTE + HistGB	0.0870
04	TabNet	0.0830

BASE RATE CONTEXT

BASE RATE (RTO) **1.9%** → OUR MODEL (PR-AUC) **0.1076**

0.019 → 0.1076 (PR-AUC)

ECONOMIC CONTEXT — THE CALLBACK

₹50,000 crore annual RTO loss in India (est.) → Stronger RTO signal 5.7× / 32× lift → Preventable loss opportunity

Even a fraction of this signal is real money back.

HONEST GAPS — BEFORE YOU ASK

#	GAP AREA	CURRENT STATE (TODAY)	DETAILS	NEXT STEP	STATUS
01	MOCK SCORER	Deployed console today	Console uses deterministic mock scorer	Swap to live scorer (one env var change)	IN PROGRESS
02	CALIBRATION	Probabilities run hot	Brier score 0.052 (not yet calibrated)	Platt / Isotonic (sigmoid) calibration planned	IN PROGRESS
03	MERKLE (FILE-MODE)	File-mode merkle in multi-process	One deliberate seam in demo deployment	Move to Postgres mode for horizontal scale	PLANNED
04	LATENCY	45–75 ms (p50) end-to-end	Depends on network + feature lookups	Target <10 ms p50 (production path)	NOT CLAIMED

LEGEND

- Data / Flow
- Validation / Cross-check
- Lookup / Reference

NOTES

- All results are time-split.
- No user history in left dataset.
- Olist cross-check uses per-customer history.
- Same pipeline, same model family, different data.

BUSINESS MODEL — WE PRICE THE OUTCOME, NOT THE SEAT

Zero-MDR compressed payment margins. **Merchant outcomes are the margin** — and every outcome here is measured in prevented RTO loss.

1) PRICING — PER DECISION

₹0.25

per scored order · **volume tiers ₹0.10–₹0.50**

SAAS TIER

₹999 / mo per merchant

risk console, rule engine & audit exports included

UNIT ECONOMICS

₹200–250 saved per prevented RTO — the ₹0.25 decision is a **roughly 1,000× payoff** on every saved parcel

VOLUME TIERS

₹0.50	₹0.25	₹0.10
< 10K orders/mo	10K – 100K	100K +

Pays for itself at 5 prevented RTOs / month.

2) WEDGE — DISTRIBUTION

13.7M

Paytm merchant device subscribers **already paying monthly** for merchant tooling

MARKET PULL

40–60% of Indian e-commerce orders are still COD — every one of them needs a risk decision before pickup

PAYTM MERCHANT APP → RISK CONSOLE → DECISION API

The risk console slots into the **same subscription motion as Paytm's Soundbox**. It's a decision layer on rails merchants already trust, billed on the statement they already pay.

Same merchant, same bill, new line item: decisions, not hardware.

3) EXPANSION — COMPOUNDING

1. **Merchant lending risk**
   RTO history doubles as a repayment signal — one data asset, two underwriting products.
2. **Agentic-commerce guardrails**
   License the bounded-agent + OC-201B cap layer to any agent spending on UPI.
3. **RTO insurance pricing**
   Per-merchant loss curves make return insurance underwritable.

THE CEILING

₹50,000 Cr annual COD RTO loss in India (est.) — every rupee prevented is a rupee of the outcome we price

Where this goes: the audit trail that makes decisions contestable today becomes the underwriting data that prices credit and insurance tomorrow.

₹ Zero-MDR made payment margin near-zero. **We price the outcome** — a share of prevented loss, not a share of trust.

WE BUILT THE PART THAT MATTERS MOST.

trust core · detector · audit trail

repo public · console live · tests green

Try it. Break it. Check the audit trail.

github.com/Neeraj-Parekh/special-parakeet

rto-trust-layer.vercel.app

Team RTO Trust Layer · Neeraj Parekh · Paytm Build for India AI Hackathon — Track 1: Merchant Growth AI